

Supplementary Material

Supplementary Methods

Optimalization of the cut-off of the size of LOH regions

In order to find the most appropriate cutoff value for the minimum length of LOH to be considered in the HRD-LOH score we investigated that which cutoff-value based HRD-LOH score would be able to discriminate the BRCA-deficient, and intact samples the best. We considered using Mann-Whitney or Kolmogorov-Smirnov. (Both nonparametric tests to compare two unpaired groups of data.) Abkevich et al. ¹ used Kolmogorov-Smirnov test, however Mann-Whitney is more sensitive to changes in the median of the groups, thus we carried out the analysis both with Kolmogorov-Smirnov test (**Supplementary Figure S1A** and Mann-Whitney test (**Supplementary Figure S1B**). In the Abkevich et al. paper the cut-off of 15 Mb was slightly arbitrarily selected, rather than selecting the cut-off with the lowest p-value (cut-off of 7 Mb). The authors argue that the rational for this selection rather than selecting the cut-off with the lowest p-value is that the 15 Mb cut-off is more sensitive to statistical noise present in the data.

We also performed Spearman rank correlation between the SNP-array-based and WXS-based HRD-LOH scores for the different cutoff minimum LOH lengths (**Supplementary Figure S1C**). Here the 14 Mb and 15 Mb cutoff-based WXS-HRD-LOH score had the highest correlation with the SNP-based HRD score. (0.700 and 0.695 respectively). This result reassured our choice of using the 15 Mb cutoff like in the SNP-array-based HRD-LOH score.

Predictive value of WXS-based HRD-sum

The predictive value of HRD-sum, measured as AUC value of the corresponding ROC curve, was 80.8% (**Supplementary Figure S2**).

The effect of WXS-based estimation of genomic scar scores on their discriminating value

In order to investigate if either genomic scar score individually deteriorates on WXS data, we compared the ROC-curves of BRCA-status predictive potential for each of the scar scores and for HRD-sum between the SNP-based and the WXS-based results. To plot the curves and test difference we used the pROC R package ². None of the WXS-based curves were significantly different from the SNP-array-based ones (DeLong's test, $n = 95$). (**Supplementary Figure S3A-D**)

Differences in the genomic scar score between BRCA-deficient and BRCA-intact samples

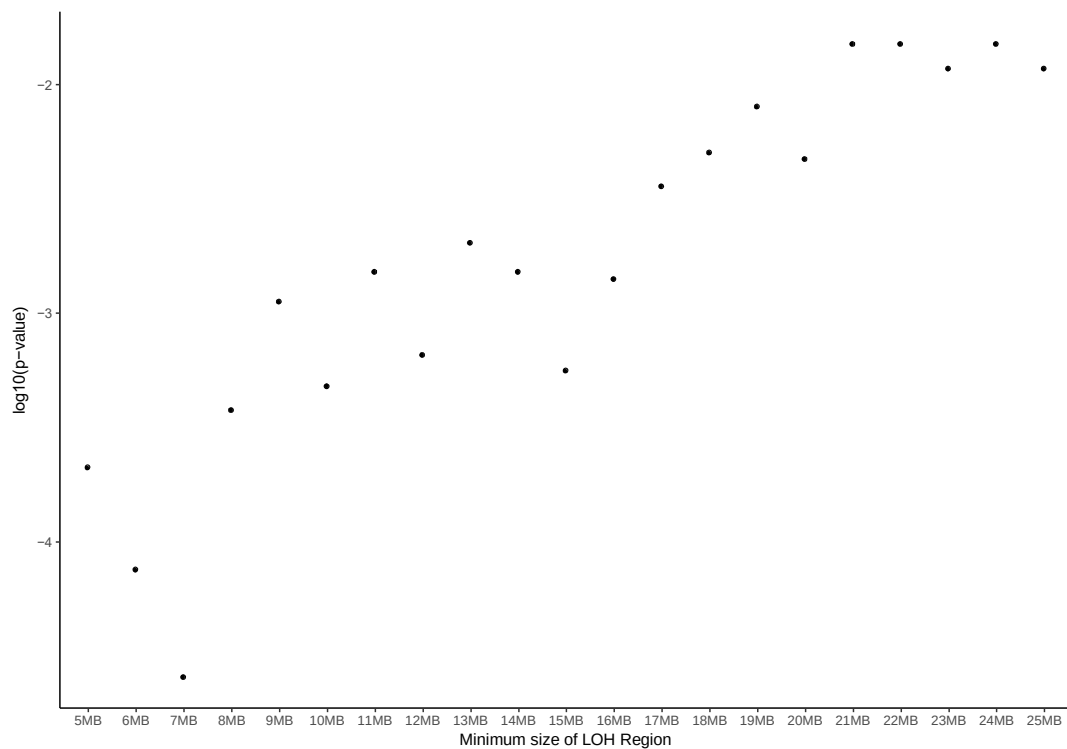
We investigated the differences between BRCA-deficient and BRCA-intact samples and all of the score were significantly elevated in the in BRCA-deficient group ($p < 0.001$, Mann-Whitney test) (**Supplementary Figure S4A-D, Supplementary Figure S5**)

So we can represent the difference of HRD-sum distribution between BRCA Intact and Deficient samples we prepared a version of Figure 2, where the y-axis represents the fraction of the samples from the BRCA Intact or Deficient group instead of the number of patients. **(Supplementary Figure S6)**

Effect of coverage on the individual genomic scar scores

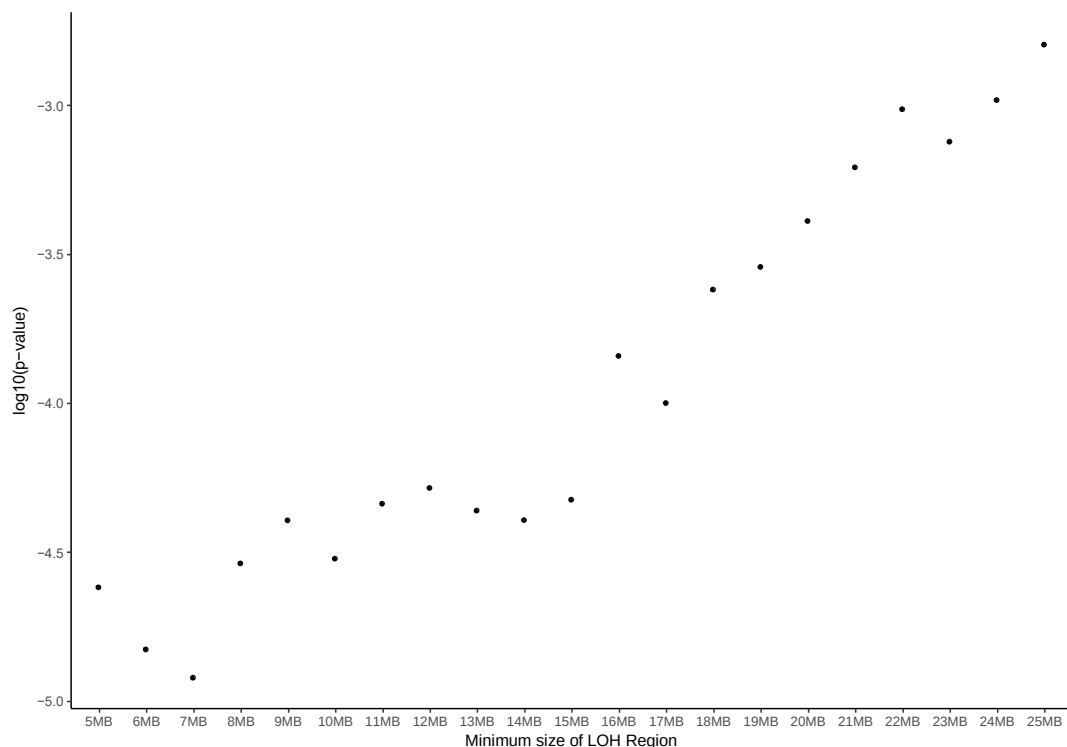
To evaluate the effect of coverage on the individual genomic scar scores, we determined the coverage of 15-15 randomly selected BRCA-deficient and BRCA-intact samples. The coverage was estimated using Mosdepth³. The Pearson correlation between the coverage and the genomic scars were low in each of the cases **(Supplementary Figure S7)**. The HRD-LOH score showed the “highest” correlation ($r = 0.067$, $r^2 = 0.0045$) with average of tumor-, normal depth. To further test the effect of coverage we simulated a reduced, 30x coverage for these samples selecting reads randomly using SAMtools for this subsampling⁴. (The original coverage of these 30 germline and tumor samples were 79x, 85x, respectively. Each of the genomic scar score showed high Pearson-correlation between the 30x (subsampled) and the original samples. **(Supplementary Figure S8)**

Supplementary Figure S1A



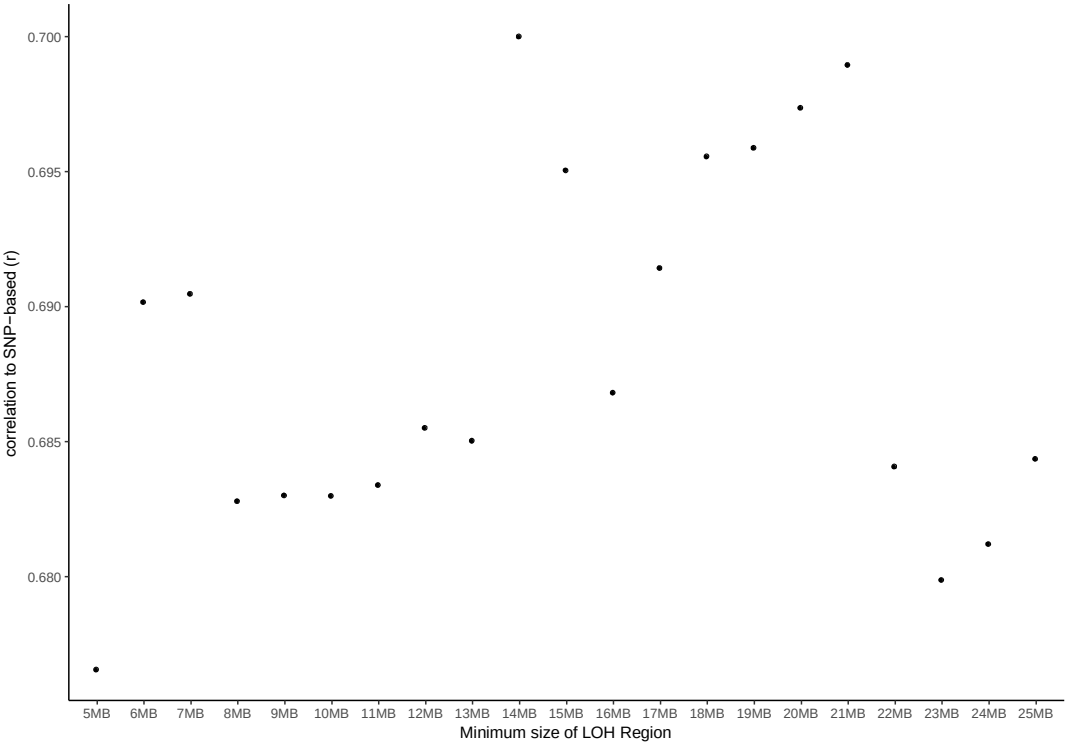
Supplementary Figure S1A: Optimization of the cut-off of the size of LOH regions. Correlation between HRD-LOH scores and BRCA1/2 deficiency was calculated for different LOH region length cut-offs. The corresponding log10(p-value) of the Kolmogorov-Smirnov test are on the y-axis. Since the original (SNP-based) 15 Mb cut off value was one of the most significant lower size limits we chose to use this in our further analysis.

Supplementary Figure S1B



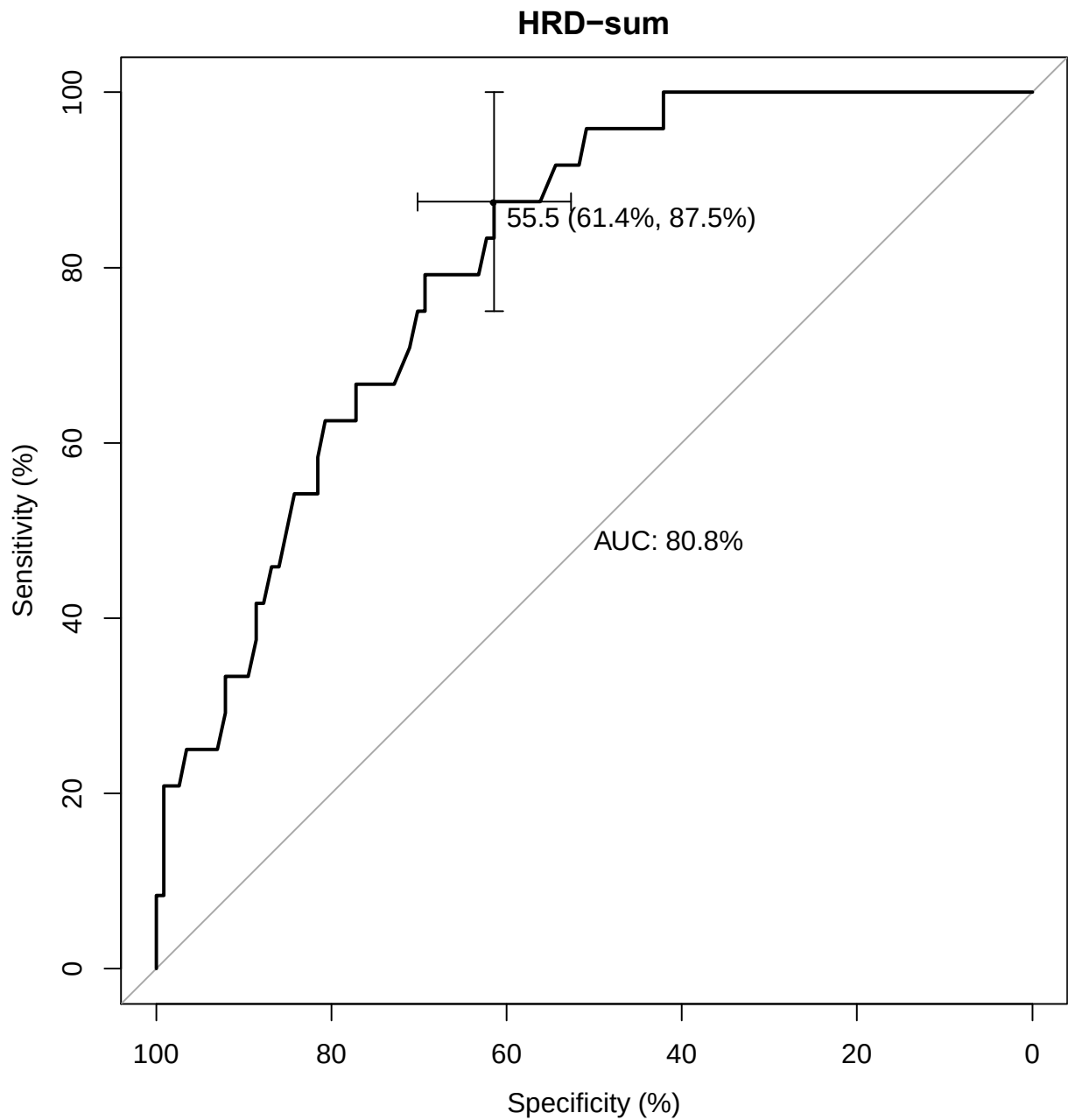
Supplementary Figure S1B: Optimization of the cut-off of the size of LOH regions. Correlation between HRD-LOH scores and BRCA1/2 deficiency was calculated for different LOH region length cut-offs. The corresponding log10(p-value) of the Mann-Whitney test are on the y-axis.

Supplementary Figure S1C



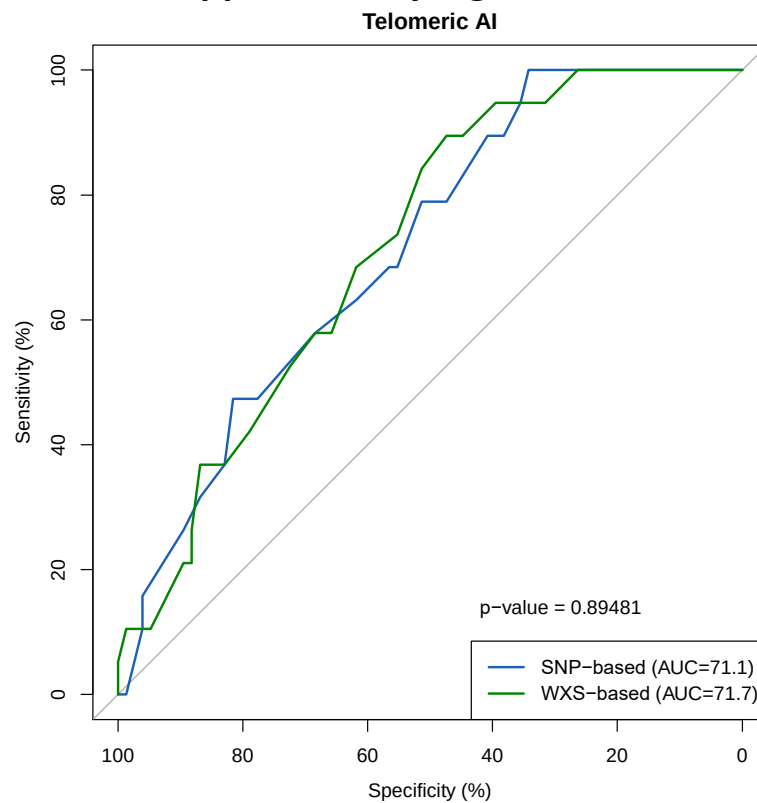
Supplementary Figure S1C: Spearman rank correlation between the SNP-array-based and WXS-based HRD-LOH scores for the different cutoff minimum LOH lengths (5 Mb to 25 Mb).

Supplementary Figure S2



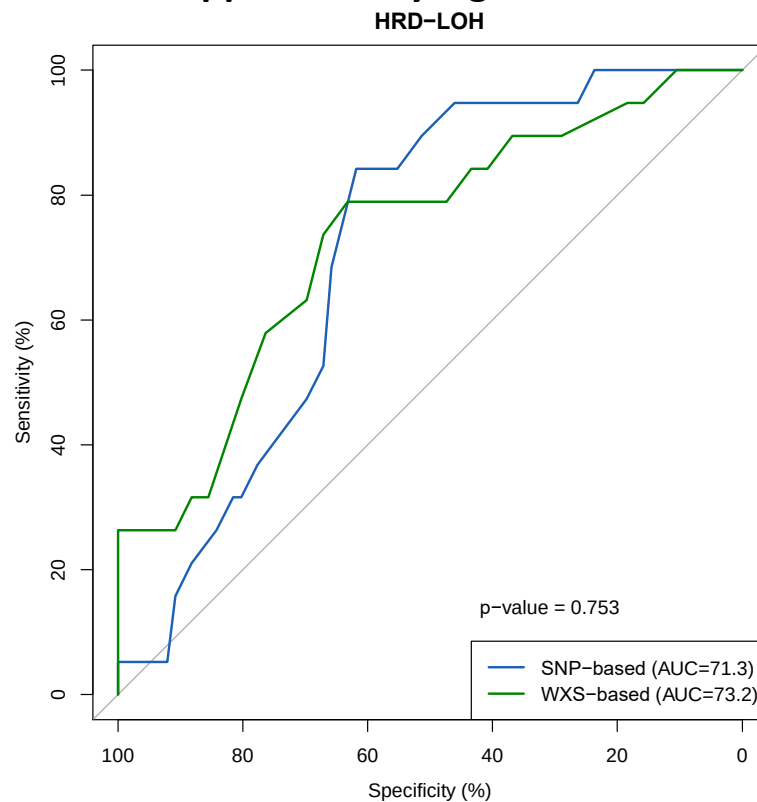
Supplementary Figure S2: ROC curve showing the ability of the WXS-based HRD-sum to predict the BRCA1/2 deficiency status in n=139 triple-negative breast cancer patients of the TCGA dataset. The error bar around the best cutoff value represents the 95% confidence interval of the sensitivity and specificity.

Supplementary Figure S3A



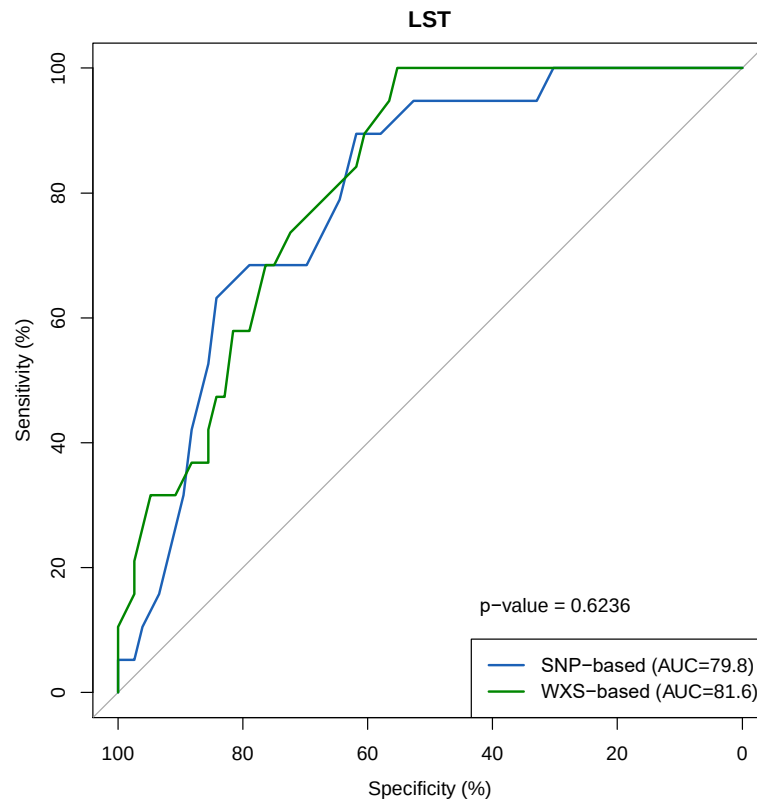
Supplementary Figure S3A: ROC curves showing the ability of the WXS-based and SNP-array based Telomeric Allelic Imbalance score to predict the BRCA1/2 deficiency status in n=95 triple-negative breast cancer patients of the TCGA dataset

Supplementary Figure S3B



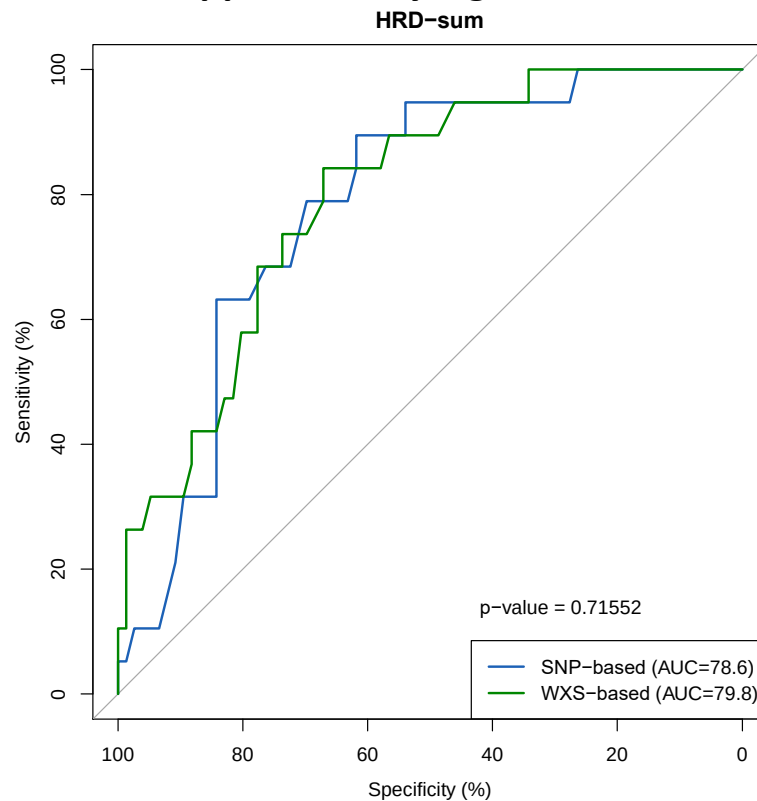
Supplementary Figure 3B: ROC curves showing the ability of the WXS-based and SNP-array based HRD-LOH score to predict the BRCA1/2 deficiency status in n=95 TNBC patients of the TCGA dataset

Supplementary Figure S3C



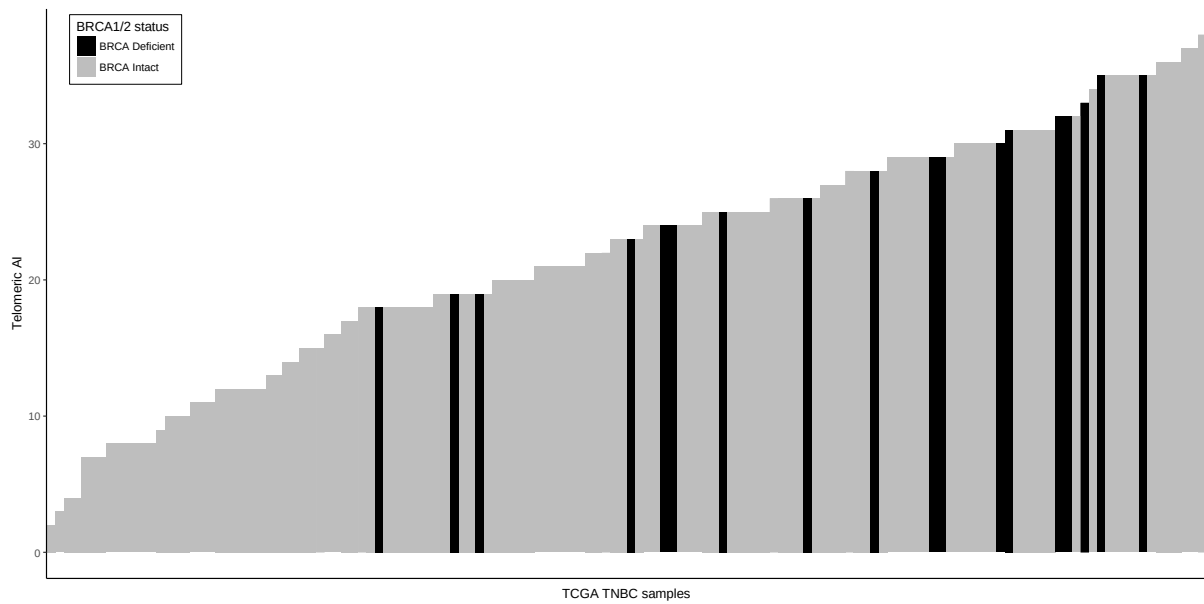
Supplementary Figure 3C: ROC curves showing the ability of the WXS-based and SNP-array based LST score to predict the BRCA1/2 deficiency status in n=95 TNBC patients of the TCGA dataset

Supplementary Figure S3D



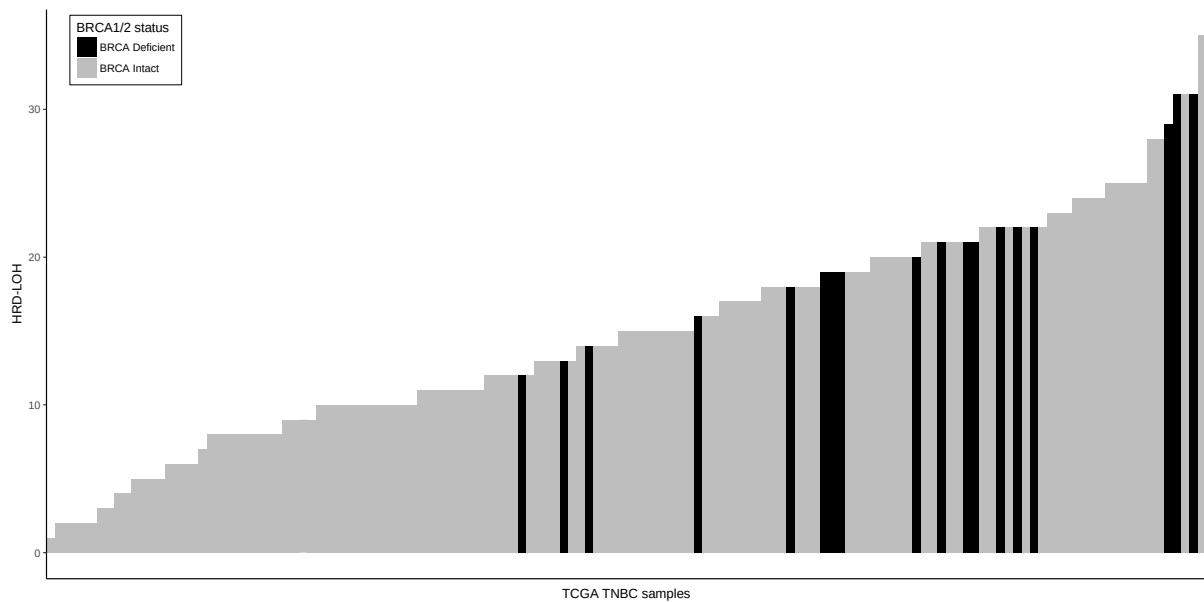
Supplementary Figure 3D: ROC curves showing the ability of the WXS-based and SNP-array based HRD-sum score to predict the BRCA1/2 deficiency status in n=95 TNBC patients of the TCGA dataset

Supplementary Figure S4A:



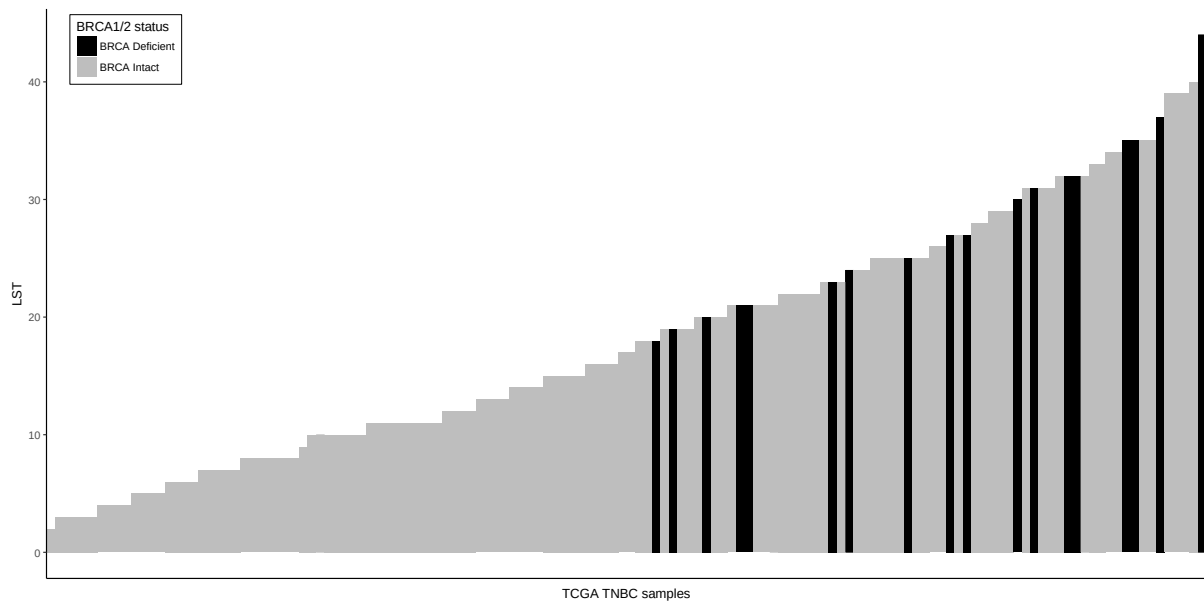
Supplementary Figure S4A: Distribution of the number of Telomeric allelic imbalances in TNBC patients (n=139), the black lines represent the BRCA1/2 deficient samples.

Supplementary Figure S4B:



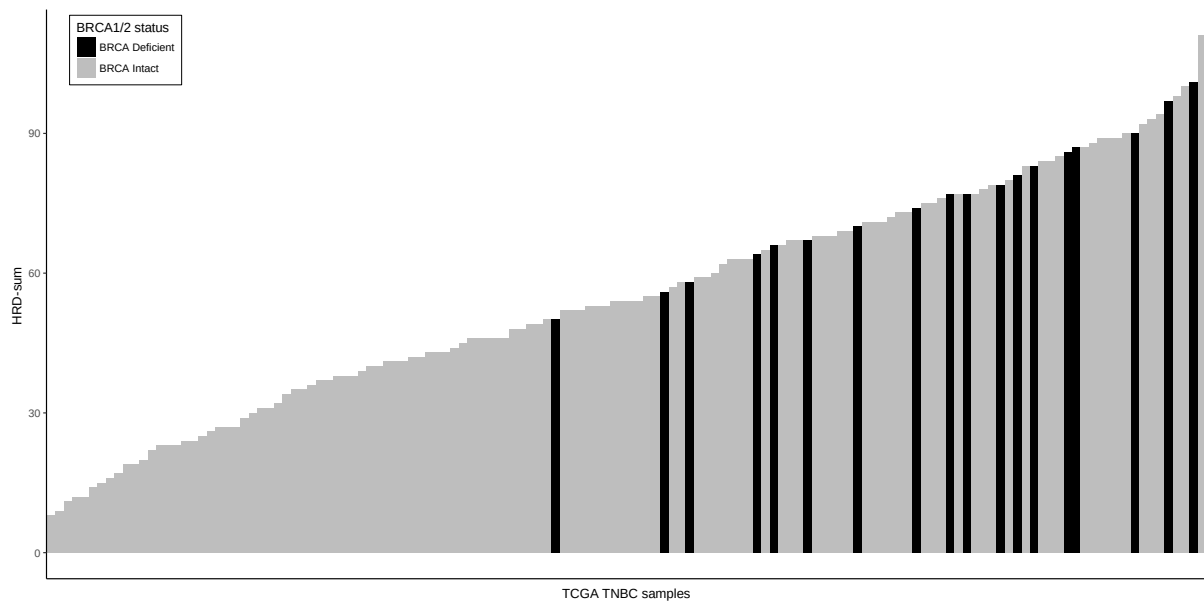
Supplementary Figure S4B: Distribution of the HRD-LOH scores in TNBC patients (n=139), the black lines represent the BRCA1/2 deficient samples.

Supplementary Figure S4C:



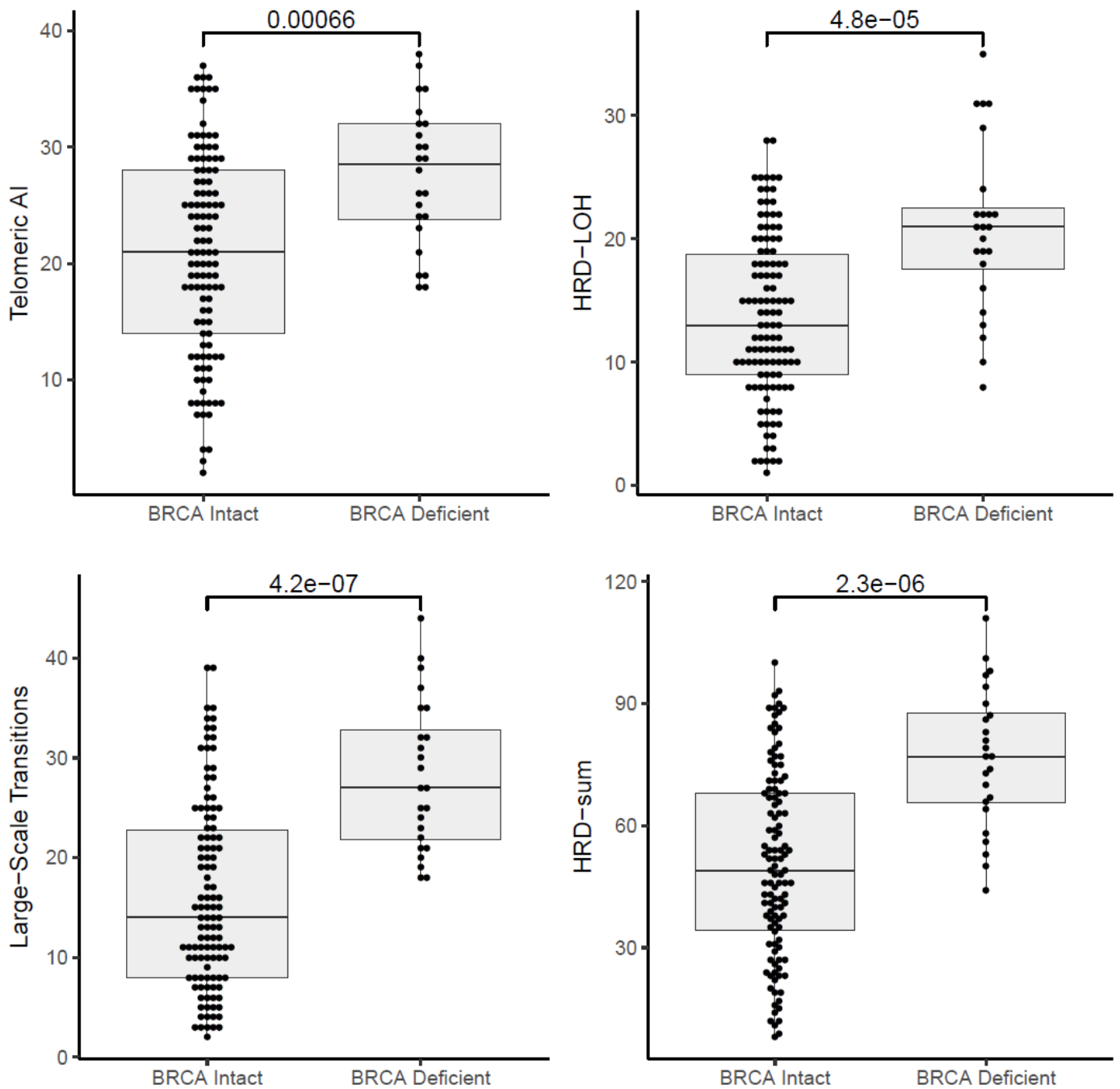
Supplementary Figure S4C: Distribution of the number of Large-Scale Transitions (LSTs) in TNBC patients (n=139), the black lines represent the BRCA1/2 deficient samples.

Supplementary Figure S4D:



Supplementary Figure S4D: Distribution of HRD-sum values in TNBC patients (n=139), the black lines represent the BRCA1/2 deficient samples.

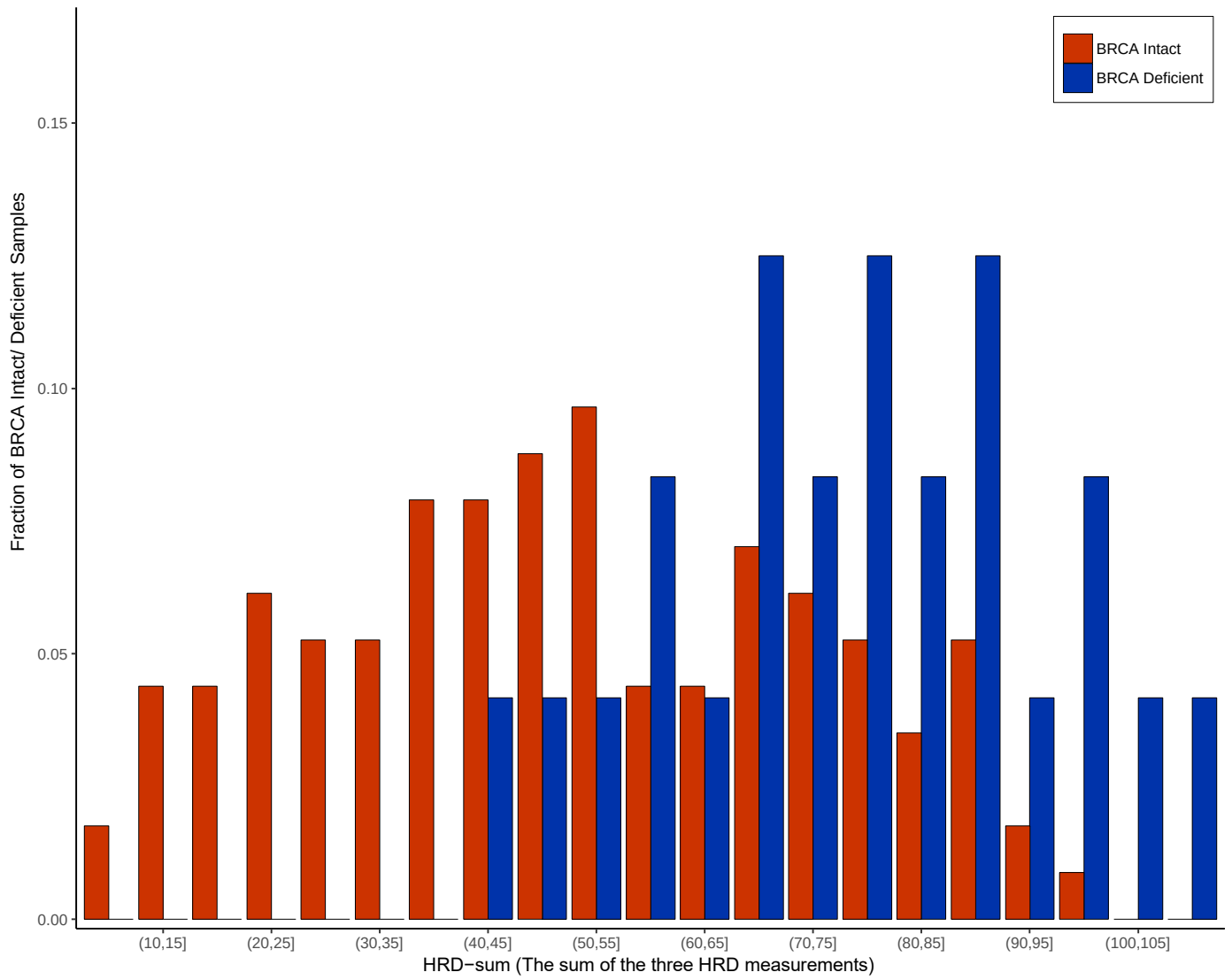
Supplementary Figure S5



Supplementary Figure S5: Differences between BRCA-deficient and BRCA-intact samples for the WXS-derived genomic scar scores (n=139.)

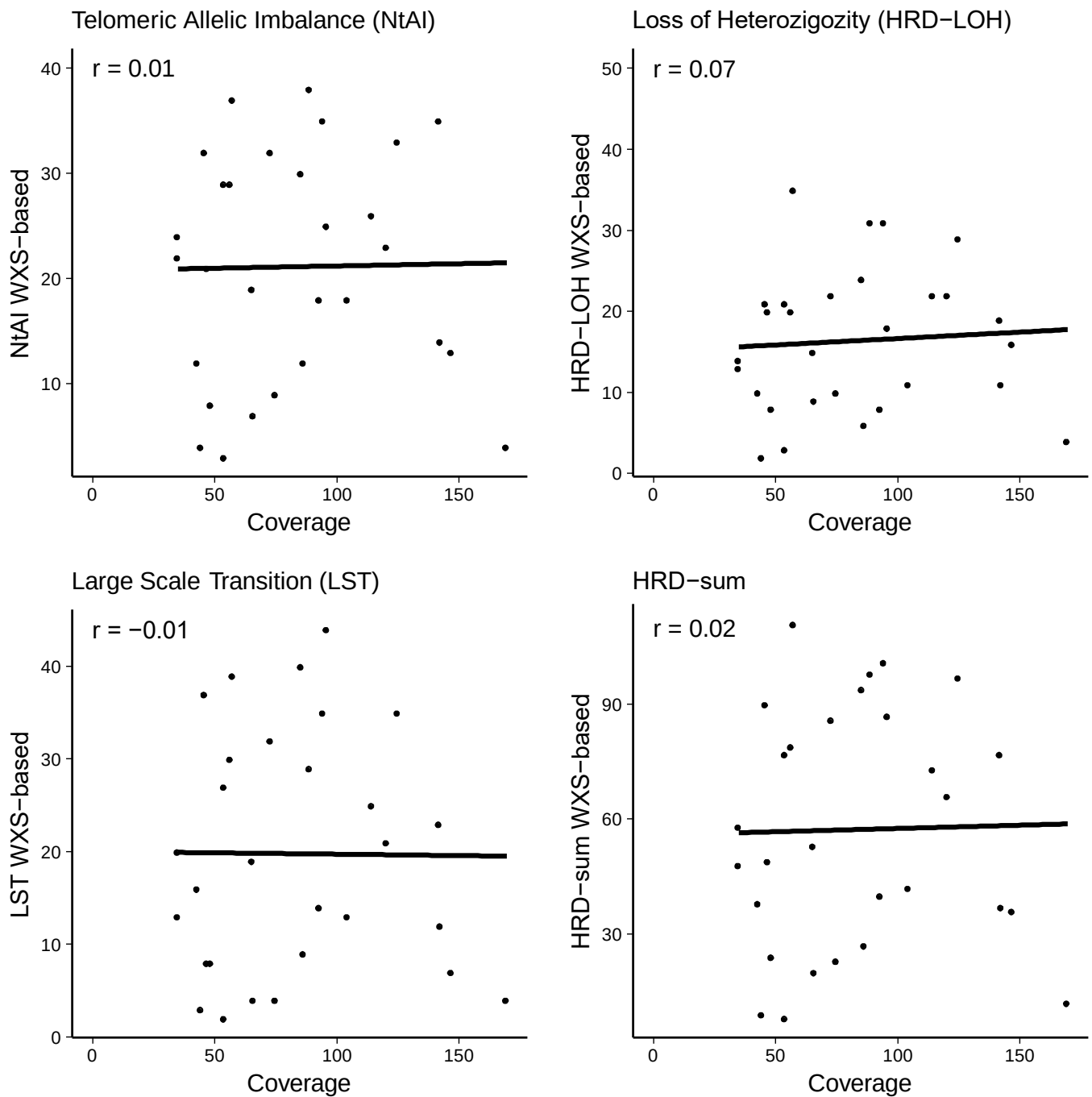
A sample was classified as BRCA-deficient if (1) there was a deep deletion of BRCA1/2, (2) a germline and a somatic mutation in BRCA1/2 with LOH, or (3) if LOH had co-occurred with promoter methylation in one of the BRCA1/2 genes.

Supplementary Figure S6



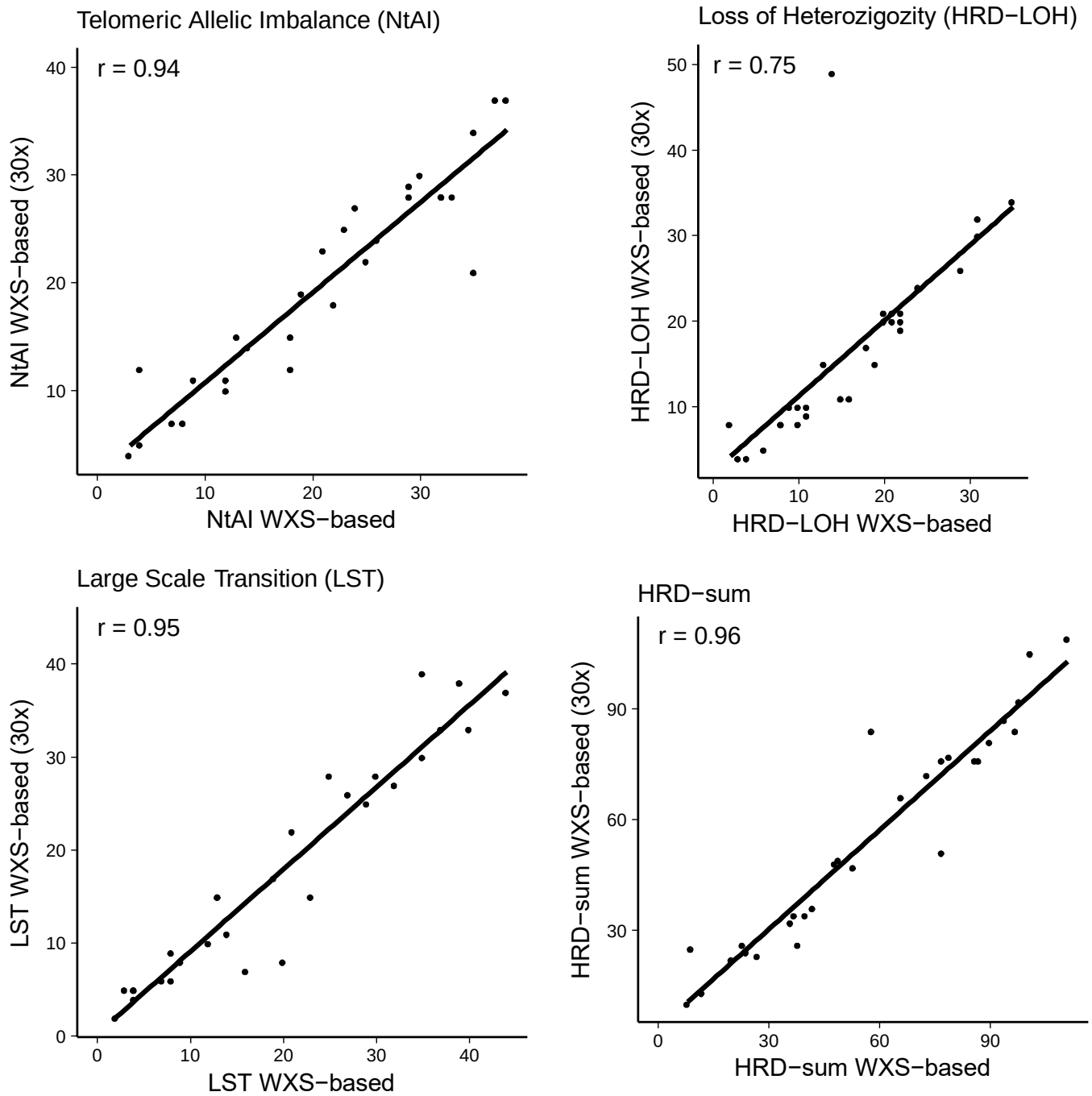
Supplementary Figure S6: The difference of HRD-sum distribution between BRCA Intact and Deficient samples represented in a modified version of Figure 2, where the y-axis represents the fraction of the samples from the BRCA Intact or Deficient group instead of the number of patients

Supplementary Figure S7



Supplementary Figure S7: The effect of coverage on the individual genomic scar scores, in 15-15 randomly selected BRCA-deficient and BRCA-intact samples. The Pearson correlation between the coverage and the genomic scars were low in each of the cases

Supplementary Figure S8



Supplementary Figure S8: The effect of lower coverage (30x) on the genomic scar scores in 15-15 randomly selected BRCA-deficient and BRCA-intact samples. Each of the genomic scar score showed high Pearson-correlation between the 30x (subsampled) and the original samples.

References

- 1 Abkevich, V. *et al.* Patterns of genomic loss of heterozygosity predict homologous recombination repair defects in epithelial ovarian cancer. *Br. J. Cancer* **107**, 1776-1782, doi:10.1038/bjc.2012.451 (2012).
- 2 Robin, X. *et al.* pROC: an open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics* **12**, 77, doi:10.1186/1471-2105-12-77 (2011).
- 3 Pedersen, B. S. & Quinlan, A. R. Mosdepth: quick coverage calculation for genomes and exomes. *Bioinformatics* **34**, 867-868, doi:10.1093/bioinformatics/btx699 (2018).
- 4 Li, H. *et al.* The Sequence Alignment/Map format and SAMtools. *Bioinformatics* **25**, 2078-2079, doi:10.1093/bioinformatics/btp352 (2009).

Supplementary Table S1

PatientID	HRD.LOH_WXS	Telomeric.AI_WXS	LST_WXS	HRDsum_WXS	HRD.LOH_SNP	Telomeric.AI_SNP	LST_SNP	HRDsum_SNP	BRCA.status	BRCA1germline	BRCA2germline	BRCA1somatic	BRCA2somatic	BRCA1_LOH	BRCA2_LOH	BRCA1methylation(beta-value)	BRCA2methylation(beta-value)
TCGA-A1-A0SK	8	18	14	40	10	17	18	45	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-A1-A0SO	31	35	35	101	23	33	29	85	BRCA1 Deficient (germline mutation and LOH)	1	0	1	0	1	0	0.03	0.01
TCGA-A1-A0SP	10	11	10	31	11	9	3	23	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-A2-A04P	22	36	31	89	21	34	28	83	BRCA Intact	0	0	0	1	1	0	NA	NA
TCGA-A2-A04Q	9	17	16	42					BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-A2-A04T	29	33	35	97	29	32	28	89	BRCA1 Deficient (germline mutation and LOH)	1	0	0	0	1	0	NA	NA
TCGA-A2-A04U	17	30	16	63	12	28	19	59	BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-A2-A0CM	9	20	17	46					BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-A2-A0D0	14	29	28	71	25	34	28	87	BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-A2-A0D2	19	31	31	81					BRCA1 Deficient (germline mutation and LOH)	1	0	0	0	1	1	NA	NA
TCGA-A2-A0ST	1	8	7	16					BRCA Intact	0	0	0	0	0	0	0.02	0.01
TCGA-A2-A0SX	12	17	10	39	31	14	14	59	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-A2-A0T0	18	27	34	79	21	22	36	79	BRCA Intact	0	1	0	1	0	0	0.02	0.01
TCGA-A2-A0T2	15	23	11	49	10	13	8	31	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-A2-A0YE	17	19	10	46	30	23	11	64	BRCA Intact	0	0	0	0	1	1	0.04	0.01
TCGA-A2-A1G6	2	7	3	12	0	0	0	0	BRCA Intact	0	0	0	0	0	0	0.02	0.01
TCGA-A2-A3XS	15	18	13	46	19	22	15	56	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-A2-A3XT	10	28	21	59	18	24	33	75	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-A2-A3XU	9	7	4	20	7	7	5	19	BRCA Intact	0	0	0	0	0	0	0.02	0.01
TCGA-A2-A3XX	8	8	3	19	4	18	25	47	BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-A7-A0DA	15	35	39	89	25	38	32	95	BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-A7-A26F	11	23	18	52	13	13	13	39	BRCA Intact	0	0	0	0	1	0	0.03	0.01
TCGA-A7-A26G	11	12	15	38					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-A7-A26I	22	21	7	50	24	29	8	61	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-A7-A4SD	15	19	19	53	17	23	13	53	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-A7-A4SE	24	27	16	67	36	28	17	81	BRCA Intact	0	0	0	0	1	1	0.04	0.01
TCGA-A7-A6VV	18	26	10	54					BRCA Intact	0	0	0	0	1	1	0.02	0.01

PatientID	HRD.LOH_WXS	Telomeric.AI_WXS	LST_WXS	HRDsum_WXS	HRD.LOH_SNP	Telomeric.AI_SNP	LST_SNP	HRDsum_SNP	BRCA.status	BRCA1germline	BRCA2germline	BRCA1somatic	BRCA2somatic	BRCA1_LOH	BRCA2_LOH	BRCA1methylation(beta-value)	BRCA2methylation(beta-value)
TCGA-A7-A6VW	16	19	21	56					BRCA1 Deficient (somatic mutation and LOH)	0	0	1	0	1	1	0.03	0.01
TCGA-A7-A6VY	22	18	24	64					BRCA1 Deficient (LOH and promoter hypermethylation)	0	0	0	0	1	1	0.46	0.01
TCGA-A8-A07O	6	12	9	27	7	10	6	23	BRCA Intact	0	0	0	0	0	1	NA	NA
TCGA-A8-A08R	23	35	35	93	29	36	27	92	BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-A8-A09X	5	24	14	43					BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-AC-A2BK	22	22	11	55	28	28	21	77	BRCA Intact	0	0	0	0	0	1	0.03	0.01
TCGA-AC-A2QH	13	22	13	48	15	18	16	49	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-AC-A2QJ	10	10	3	23	2	7	5	14	BRCA Intact	0	0	0	0	0	1	0.02	0.01
TCGA-AC-A6IW	11	24	10	45					BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-AC-A7VC	7	10	5	22					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-AN-A04D	11	21	26	58	13	21	27	61	BRCA Intact	0	0	0	0	0	1	NA	NA
TCGA-AN-A0AL	22	23	21	66	23	25	30	78	BRCA1 Deficient (germline mutation and LOH)	1	0	0	1	1	0	NA	NA
TCGA-AN-A0AR	25	28	32	85	32	29	30	91	BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-AN-A0AT	25	36	39	100	25	27	23	75	BRCA Intact	0	0	0	1	1	0	NA	NA
TCGA-AN-A0G0	11	21	11	43	10	40	19	69	BRCA Intact	0	0	0	0	0	1	NA	NA
TCGA-AN-A0XU	24	30	40	94	22	31	31	84	BRCA Intact	0	0	1	0	1	0	0.75	0.01
TCGA-AO-A0J2	24	30	35	89	26	33	33	92	BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-AO-A0J4	10	21	22	53	20	23	20	63	BRCA Intact	0	0	1	0	1	0	NA	NA
TCGA-AO-A0J6	22	26	25	73	22	27	32	81	BRCA Intact	1	0	0	0	1	1	NA	NA
TCGA-AO-A124	31	38	29	98	30	38	23	91	BRCA Intact	1	0	0	1	1	1	0.02	0.01
TCGA-AO-A128	23	35	25	83					BRCA Intact	0	0	0	0	1	0	0.03	0.01
TCGA-AO-A1KR	8	18	18	44	11	21	13	45	BRCA Intact	0	0	1	0	1	1	0.02	0.01
TCGA-AQ-A04J	15	20	24	59					BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-AQ-A54N	20	21	8	49	19	23	4	46	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-AR-A0TS	13	30	25	68					BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-AR-A0TU	35	37	39	111	41	35	38	114	BRCA Intact	0	0	0	0	1	1	0.64	0.01

PatientID	HRD.LOH_WXS	Telomeric.AI_WXS	LST_WXS	HRDsum_WXS	HRD.LOH_SNP	Telomeric.AI_SNP	LST_SNP	HRDsum_SNP	BRCA.status	BRCA1germline	BRCA2germline	BRCA1somatic	BRCA2somatic	BRCA1_LOH	BRCA2_LOH	BRCA1methylation(beta-value)	BRCA2methylation(beta-value)
TCGA-AR-A0U1	25	34	31	90	36	29	32	97	BRCA Intact	0	0	0	0	0	1	NA	NA
TCGA-AR-A1AI	18	26	21	65	28	32	28	88	BRCA Intact	0	0	0	1	1	0	0.02	0.01
TCGA-AR-A1AQ	10	20	8	38					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-AR-A1AR	16	13	7	36	11	13	7	31	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-AR-A1AY	11	14	12	37	21	12	10	43	BRCA Intact	0	0	0	0	1	1	0.04	0.01
TCGA-AR-A256	20	30	21	71	21	29	17	67	BRCA Intact	0	0	0	0	1	0	0.03	0.01
TCGA-AR-A2LH	6	11	6	23					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-AR-A2LR	20	26	22	68	19	29	12	60	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-AR-A5QQ	5	7	5	17					BRCA Intact	0	0	0	0	0	1	0.02	0.01
TCGA-B6-A3ZX	8	18	11	37					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-B6-A400	25	28	23	76	25	32	24	81	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-B6-A402	11	25	12	48	8	24	10	42	BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-B6-A409	12	12	10	34					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-BH-A0AV	21	27	25	73					BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-BH-A0B3	6	10	16	32					BRCA Intact	0	0	0	0	0	0	0.04	0.01
TCGA-BH-A0B9	11	18	13	42	16	16	12	44	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-BH-A0BG	20	35	12	67	22	28	20	70	BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-BH-A0BL	19	35	23	77	25	23	23	71	BRCA1 Deficient (germline mutation and LOH)	1	0	0	0	1	1	NA	NA
TCGA-BH-A0BW	15	25	22	62	24	20	21	65	BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-BH-A0E0	8	16	8	32	27	19	9	55	BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-BH-A0RX	2	12	11	25					BRCA Intact	0	0	0	0	0	0	0.03	0.01
TCGA-BH-A0WA	19	26	25	70	25	34	30	89	BRCA1 Deficient (somatic mutation and LOH)	0	0	1	0	1	1	0.05	0.01
TCGA-BH-A18G	4	4	4	12	2	4	5	11	BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-BH-A18V	28	31	33	92	37	34	27	98	BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-BH-A1F6	12	29	14	55	19	17	20	56	BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-BH-A1FC	10	20	11	41	16	32	30	78	BRCA Intact	0	0	0	0	0	1	0.02	0.01
TCGA-BH-A42U	10	11	6	27	0	0	0	0	BRCA Intact	0	0	0	0	0	1	0.02	0.01

PatientID	HRD.LOH_WXS	Telomeric.AI_WXS	LST_WXS	HRDsum_WXS	HRD.LOH_SNP	Telomeric.AI_SNP	LST_SNP	HRDsum_SNP	BRCA.status	BRCA1germline	BRCA2germline	BRCA1somatic	BRCA2somatic	BRCA1_LOH	BRCA2_LOH	BRCA1methylation(beta-value)	BRCA2methylation(beta-value)
TCGA-C8-A12V	5	15	7	27					BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-C8-A131	2	8	4	14					BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-C8-A1HJ	25	37	26	88	22	27	21	70	BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-C8-A26X	10	18	12	40					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-C8-A27B	21	31	25	77	29	29	19	77	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-C8-A3M7	2	4	3	9	4	3	4	11	BRCA Intact	0	0	0	0	1	0	0.03	0.01
TCGA-D8-A13Z	16	31	28	75					BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-D8-A142	12	31	20	63	36	26	17	79	BRCA Intact	0	0	0	0	0	1	NA	NA
TCGA-D8-A143	15	16	10	41	30	25	24	79	BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-D8-A147	18	25	44	87	23	28	35	86	BRCA1 Deficient (germline mutation and LOH)	1	0	0	0	1	1	NA	NA
TCGA-D8-A1JF	6	15	8	29	16	19	18	53	BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-D8-A1JG	17	18	11	46					BRCA Intact	0	0	0	0	1	1	0.03	0.01
TCGA-D8-A1JL	12	28	27	67	31	34	32	97	BRCA1 Deficient (somatic mutation and LOH)	0	0	0	0	1	1	0.3	0.01
TCGA-D8-A1XK	18	29	22	69	26	30	26	82	BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-D8-A1XQ	22	32	32	86	19	36	29	84	BRCA1 Deficient (germline mutation and LOH)	1	0	0	0	1	0	0.02	0.01
TCGA-D8-A27F	31	24	19	74	31	29	24	84	BRCA1 Deficient (somatic mutation and LOH)	0	0	0	0	1	1	0.55	0.01
TCGA-D8-A27M	14	24	20	58	28	20	24	72	BRCA1 Deficient (somatic mutation and LOH)	0	0	1	0	1	1	0.03	0.01
TCGA-E2-A14N	21	30	32	83					BRCA1 Deficient (germline mutation and LOH)	1	0	0	0	1	0	0.03	0.01
TCGA-E2-A14R	21	36	27	84	22	34	32	88	BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-E2-A14X	6	18	6	30	6	25	29	60	BRCA Intact	0	0	0	0	1	0	NA	NA
TCGA-E2-A150	15	20	19	54	20	23	16	59	BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-E2-A158	17	29	29	75	21	27	35	83	BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-E2-A159	10	10	6	26					BRCA Intact	0	0	0	0	1	1	NA	NA
TCGA-E2-A1AZ	14	24	14	52	16	31	25	72	BRCA Intact	0	0	0	0	1	1	0.02	0.01

PatientID	HRD.LOH_WXS	Telomeric.AI_WXS	LST_WXS	HRDsum_WXS	HRD.LOH_SNP	Telomeric.AI_SNP	LST_SNP	HRDsum_SNP	BRCA.status	BRCA1germline	BRCA2germline	BRCA1somatic	BRCA2somatic	BRCA1_LOH	BRCA2_LOH	BRCA1methylation(beta-value)	BRCA2methylation(beta-value)
TCGA-E2-A1B6	8	19	8	35					BRCA Intact	0	0	0	0	1	0	0.03	0.01
TCGA-E2-A1L7	21	25	32	78	15	29	31	75	BRCA Intact	0	0	0	0	1	1	0.06	0.01
TCGA-E2-A1LG	10	9	4	23	8	9	2	19	BRCA Intact	0	0	0	0	0	0	0.02	0.01
TCGA-E2-A1LH	18	31	31	80					BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-E2-A1LI	14	24	15	53	32	30	27	89	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-E2-A1LL	13	19	22	54	12	23	23	58	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-E2-A574	19	23	15	57	18	24	17	59	BRCA Intact	0	0	0	0	0	0	0.02	0.01
TCGA-E9-A5FL	22	26	23	71	23	27	28	78	BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-EW-A1OV	15	15	11	41					BRCA Intact	0	0	0	0	1	0	0.01	0.01
TCGA-EW-A1P1	2	8	5	15	0	0	0	0	BRCA Intact	0	0	0	0	0	0	0.03	0.01
TCGA-EW-A1P4	20	24	25	69	25	28	37	90	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-EW-A1P7	5	12	7	24					BRCA Intact	0	0	0	0	0	1	0.01	0.01
TCGA-EW-A1P8	23	29	8	60	20	30	9	59	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-EW-A1PH	10	25	19	54	12	17	15	44	BRCA Intact	0	0	0	0	0	1	0.02	0.01
TCGA-EW-A3U0	19	35	33	87	29	34	24	87	BRCA Intact	0	0	0	0	1	1	0.04	0.01
TCGA-GI-A2C9	21	32	37	90	26	31	33	90	BRCA1 Deficient (somatic mutation and LOH)	0	0	0	0	1	1	0.68	0.01
TCGA-GM-A2DB	8	14	13	35					BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-GM-A2DD	3	13	15	31					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-GM-A2DF	13	19	18	50					BRCA1 Deficient (somatic mutation and LOH)	0	0	0	0	1	1	0.19	0.01
TCGA-GM-A2DH	12	25	29	66					BRCA Intact	0	0	0	0	0	0	0.01	0.01
TCGA-GM-A2DI	8	8	3	19					BRCA Intact	0	0	0	0	0	0	0.02	0.01
TCGA-GM-A3XL	20	29	30	79	24	29	32	85	BRCA1 Deficient (somatic mutation and LOH)	0	0	0	0	1	0	0.73	0.01
TCGA-HN-A2NL	28	32	24	84	25	33	24	82	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-LL-A441	13	22	17	52					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-LL-A5YO	14	21	11	46					BRCA Intact	0	0	0	0	1	0	0.09	0.01
TCGA-LL-A73Y	24	28	20	72					BRCA Intact	0	0	0	1	1	0	0.03	0.01

PatientID	HRD.LOH_WXS	Telomeric.AI_WXS	LST_WXS	HRDsum_WXS	HRD.LOH_SNP	Telomeric.AI_SNP	LST_SNP	HRDsum_SNP	BRCA.status	BRCA1germline	BRCA2germline	BRCA1somatic	BRCA2somatic	BRCA1_LOH	BRCA2_LOH	BRCA1methylation(beta-value)	BRCA2methylation(beta-value)
TCGA-LL-A740	3	3	2	8	3	3	3	9	BRCA Intact	0	0	0	0	0	0	0.02	0.01
TCGA-OL-A5D6	8	8	8	24	7	6	7	20	BRCA Intact	0	0	0	0	1	0	0.02	0.01
TCGA-OL-A5D7	17	25	21	63	20	25	14	59	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-OL-A5RW	21	29	27	77	22	25	31	78	BRCA1 Deficient (somatic mutation and LOH)	0	0	0	0	1	0	0.85	0.01
TCGA-OL-A66I	9	19	15	43	16	21	20	57	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-OL-A66P	10	12	16	38	6	8	20	34	BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-OL-A6VO	19	29	20	68					BRCA Intact	0	0	0	0	1	1	0.02	0.01
TCGA-OL-A97C	4	2	5	11					BRCA Intact	0	0	0	0	0	0	0.03	0.01
TCGA-S3-AA10	18	25	34	77					BRCA Intact	0	0	0	0	1	1	0.03	0.03

Supplementary Table S1: Detailed table of the BRCA1/2 mutation status of the TCGA TNBC samples (germline, somatic mutations, loss-of-heterozygosity state and methylation), the SNP-array and the WXS-based genomic scar scores.

BRIEF COMMUNICATION OPEN

Migrating the SNP array-based homologous recombination deficiency measures to next generation sequencing data of breast cancer

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The first genomic scar-based homologous recombination deficiency (HRD) measures were produced using SNP arrays. As array-based technology has been largely replaced by next generation sequencing approaches, it has become important to develop algorithms that derive the same type of genomic scar scores from next generation sequencing (whole exome “WXS”, whole genome “WGS”) data. In order to perform this analysis, we introduce here the scarHRD R package and show that using this method the SNP array-based and next generation sequencing-based derivation of HRD scores show good correlation (Pearson correlation between 0.73 and 0.87 depending on the actual HRD measure) and that the NGS-based HRD scores distinguish similarly well between BRCA mutant and BRCA wild-type cases in a cohort of triple-negative breast cancer patients of the TCGA data set.

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INTRODUCTION

Reliable quantification of homologous recombination deficiency of human tumor biopsies, especially in the case of ovarian and breast cancer, is expected to identify patients that are particularly sensitive to platinum or PARP inhibitor-based therapy.¹ Before the widespread introduction of next generation sequencing (NGS) to characterize tumor biopsies, SNP arrays were used to identify large-scale genomic aberrations associated with homologous recombination deficiency, often induced by the loss of BRCA1 or BRCA2 function. Three such measures were identified: telomeric allelic imbalance (HRD-TAI score),² loss of heterozygosity profiles (HRD-LOH score),³ and large-scale state transitions (HRD-LST score).⁴ These three measures have also been combined into a single summary measure of HR deficiency.⁵ The HRD-LOH score has also become an integral part of a recently published, whole-genome sequencing-based measure of homologous recombination deficiency, HRDetect.⁶ These measures, along with functional assays,⁷ showed promise to identify HR-deficient cases and thus predict response to platinum or PARP inhibitor therapy.^{2,8,9} Since NGS has become the main genomic characterization method of cancer biopsies, it has become essential to migrate the SNP array-based methodology to NGS-based platforms.

TCGA breast cancer biopsies have been both SNP array profiled and subjected to NGS allowing a direct comparison.⁸

RESULTS AND DISCUSSION

We found good correlation between the SNP array-based and NGS-based HRD scores (Fig. 1). When comparing the results of the scarHRD R package to SNP array-based measurements, we found the following Pearson correlation coefficients: number of telomeric allelic imbalances (NtAI): $r = 0.84$ ($R^2 = 0.70$, adjusted $R^2 = 0.70$, $p < 2.2e-16$), large-scale transition (LST) $r = 0.79$ ($R^2 = 0.62$, adjusted $R^2 = 0.62$, $p < 2.2e-16$) loss of heterozygosity (HRD-LOH) $r = 0.73$ ($R^2 = 0.53$, adjusted $R^2 = 0.52$, $p < 2.2e-16$). These three measures are often combined for diagnostic purposes⁵ and in HRDetect.⁶ Therefore, we also compared the sum of the three scores across the two platforms (HRD sum): $r = 0.87$ ($R^2 = 0.75$, adjusted $R^2 = 0.75$, $p < 2.2e-16$) (Fig. 1). The artificial reduction of coverage to 30× did not affect this correlation (Supplementary Material, Figure S7–S8). The BRCA1/2-mutated samples showed significantly higher NGS-based HRD-sum values (Fig. 2, Supplementary Figure S6). The predictive value of HRD-sum, measured as AUC value of the corresponding ROC curve, was 80.8% (Supplementary Figure S2).

There was no significant difference in SNP versus WXS-based estimation of tAI, LST, and HRD-sum, but the number of LOH events were significantly lower in the WXS-based estimation ($p = 0.012$, Kolmogorov–Smirnov test). This could be attributed to differences in segmentation algorithm (the more segmented the WXS data is the lower number of LOHs that are called) or to low sample quality, coverage. However, when comparing the ROC curves for BRCA1/2 status of the SNP-based and WXS-based HRD-

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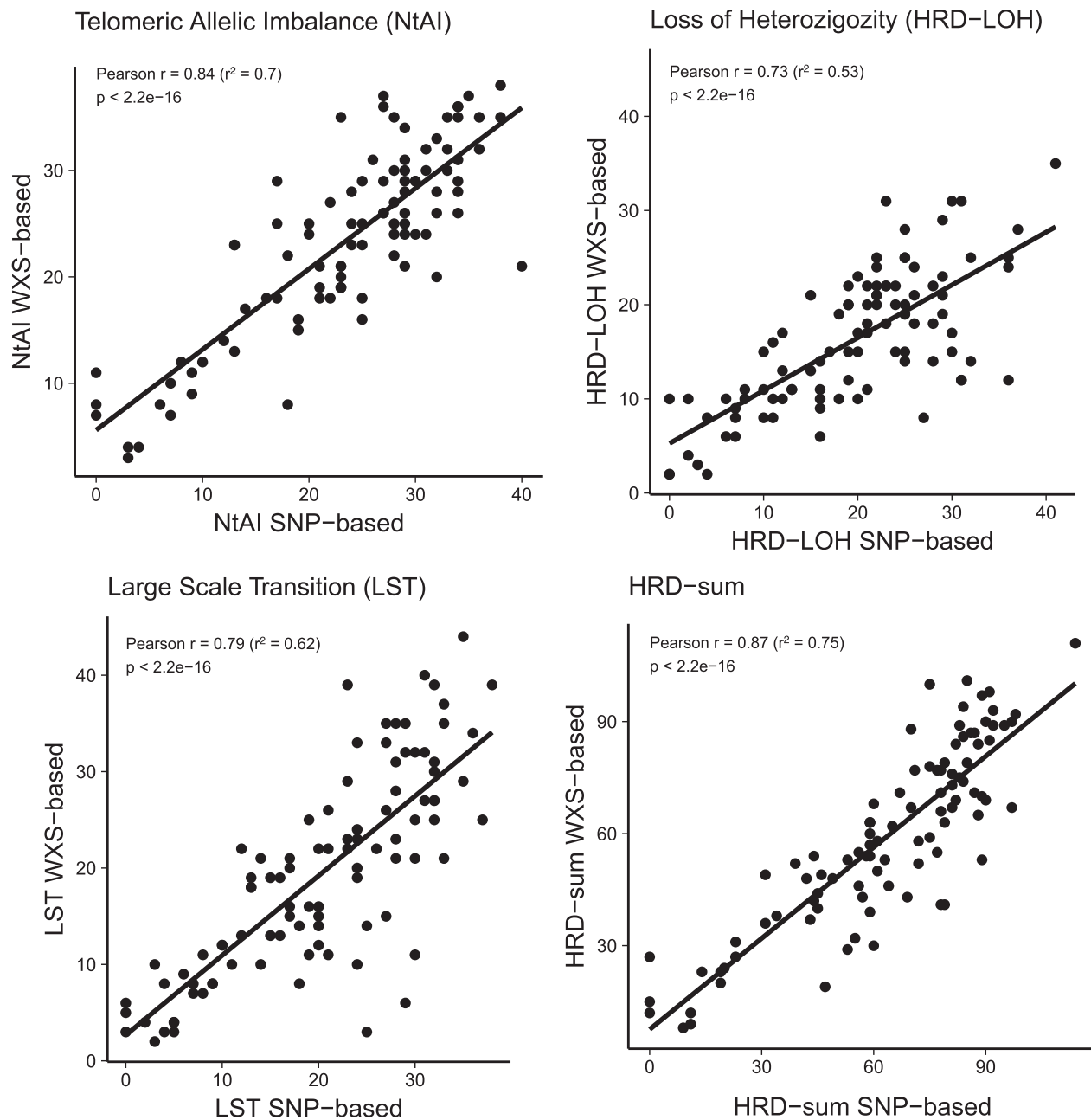


Fig. 1 Correlation between Affymetrix SNP 6.0 array-based and whole exome sequencing-based measurements of homologous recombination deficiency (telomeric allelic imbalance, loss of heterozygosity, large-scale transitions, and the sum of these estimates)

score, there was no significant difference between the SNP array-based and NGS-based methods. (Supplementary Figure S3).

According to our expectations and previous results the BRCA1/2-deficient cases showed higher values for each of the four scores (Supplementary Figure S4-S5).

The sum of the three HRD scores showed good correlation across the two platforms. Thus in more advanced NGS-based HRD deficiency measures such as HRDetect, the SNP array-based step could be replaced by an NGS-based estimate of the HR deficiency scores.

Brief description of the methods

Based on receptor status determined by immunohistochemistry, 139 paired tumor and normal samples of the TCGA breast cancer cohort could be classified as triple-negative breast cancer. From

these patients 95 had Affymetrix SNP 6.0 array-based HRD estimates (LOH, TAI, LST), previously published by our group.¹⁰ In this publication we present the scarHRD R package (<https://github.com/sztup/scarHRD>) which estimates the level of the three HR deficiency measures using NGS data.

A sample's LOH score is the total number of LOH regions across the entire genome that are larger than 15 Mb but do not cover whole chromosomes. In the original publication this 15 Mb lower limit for LOH was determined by comparing SNP array profiles between BRCA mutant and BRCA wild-type cases.³ We performed a similar analysis using NGS data and found that the original 15 Mb cutoff performed best in this case as well (Supplementary Figure S1).

The LST is defined as a chromosomal break between adjacent regions of at least 10 Mb, with a distance between them not larger than 3 Mb.

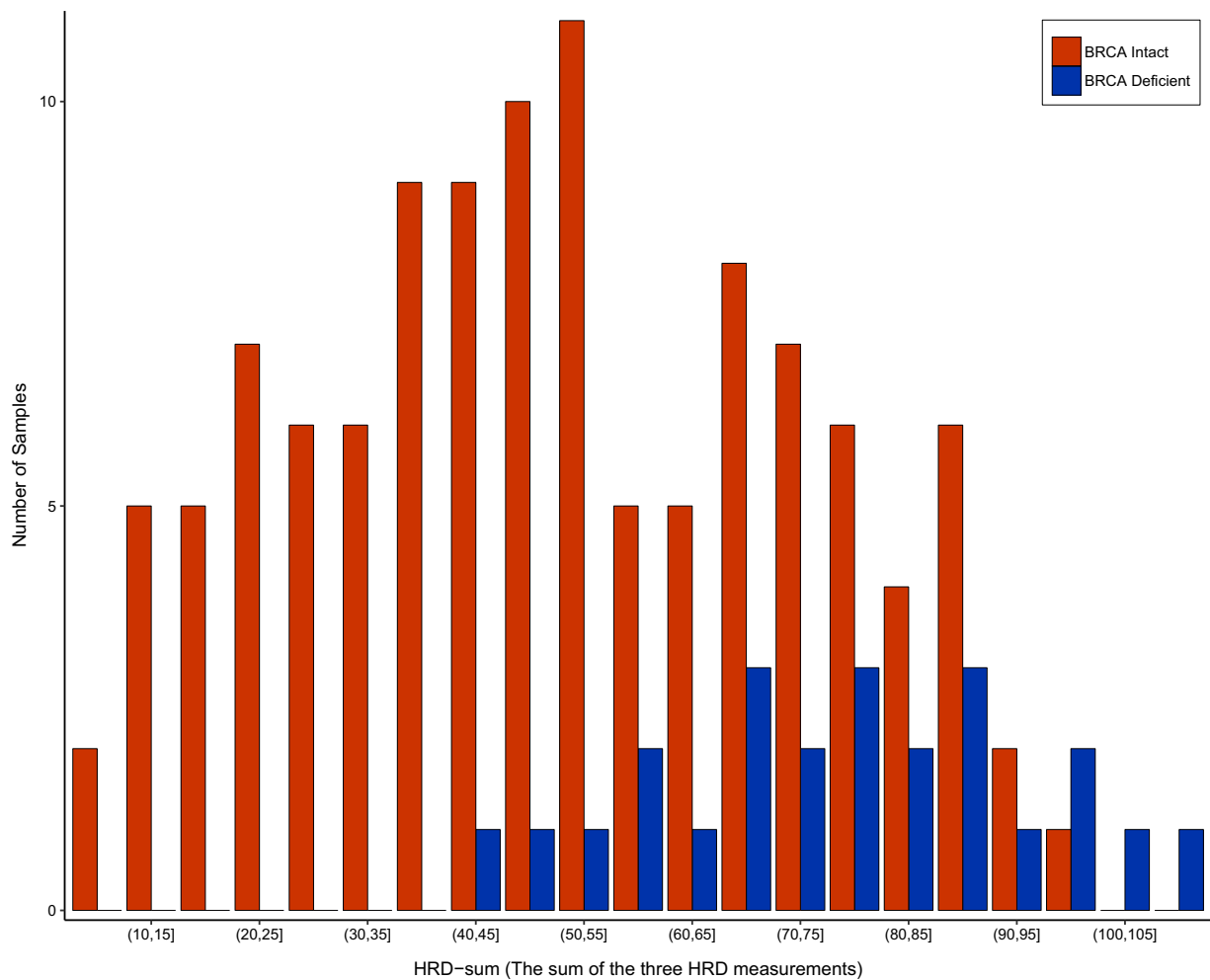


Fig. 2 Distribution of HRD-sum values in BRCA1/2 deficient and in BRCA1/2 intact triple-negative breast cancer samples from TCGA. HRD-sum values were determined with the scarHRD R package

The number of telomeric allelic imbalances is the number of AIs (the unequal contribution of parental allele sequences with or without changes in the overall copy number of the region) that extend to the telomeric end of a chromosome.

Allele-specific copy number estimation is a crucial part of estimating HR deficiency. As previously shown, allele-specific copy number estimation from NGS data performed using the Sequenza R package show high agreement with SNP array-based copy number profiles.¹¹ The scarHRD package is, therefore, able to use Sequenza preprocessed files as well as other allele-specific segmentation files in the same format.

As it has been previously shown that in ovarian cancer the sum of the genomic scar scores is elevated in BRCA-deficient cancers,⁵ an additional aim of our study was to compare the unweighted numeric sum of LOH, tAI, and LST, called here HRD-sum, to the BRCA1/2 status of the patients. A sample was classified as BRCA-deficient if (1) there was a deep deletion of BRCA1/2, (2) a germline and a somatic mutation in BRCA1/2 with LOH, or (3) if LOH had co-occurred with promoter methylation in one of the *BRCA1/2* genes. The somatic mutation status (mutations with likely pathogenic function) and methylation data was acquired from the TCGA data portal. The germline mutation status was determined using HaploTypeCaller, and was annotated with Intervar,¹² likely pathogenic mutations and frameshift insertion/deletion with unknown significance were used in our analysis. LOH was determined using Sequenza's allele-specific segmentation results (Supplementary Table S1).

Data availability

The data sets generated during the current study are available from the corresponding author on reasonable request.

Code availability

The code/algorithm for performing the experiments is available for download at <https://github.com/sztup/scarHRD>.

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AUTHOR CONTRIBUTIONS

Conception, design, writing, and review of the manuscript: Zs.S., M.D., M.K., L.R., I.C., F. F., N.J.B., A.C.E., A.S., and Zo.S. Development of methodology: Zs.S., M.D., M.K. Analysis and interpretation of data: Zs.S., M.D., M.K., Zo.S.

ADDITIONAL INFORMATION

Supplementary information accompanies the paper on the *npj Breast Cancer* website (<https://doi.org/10.1038/s41523-018-0066-6>).

Competing interests: N.J.B., A.C.E., and Zo.S are listed as co-inventors on a patent on telomeric allelic imbalance, which is owned by Children's Hospital Boston and licensed to Myriad Genetics. The remaining authors declare no competing interests.

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